Instructor Guide: Lesson 1 — Data Management and Analysis in Excel

Duration: 1 hour

Topic-by-Topic Teaching Notes

Spreadsheet Design for Technical Data

Open the lesson by showing a deliberately bad version of the OrbitCo tickets sheet — merged headers, a "Total" row pasted four rows from the top, three rows colored red with no severity column to explain why. Ask the class to list what is wrong before you say anything. You should hear at least three of: merged cells, embedded total, color-as-data, inconsistent date formats. Use their own list as the agenda for the section. The students who already know most of these things will still learn the vocabulary to name them, which is what they need when a reviewer later asks them to justify a structural choice.

The point to land is that layout decisions are reliability decisions. The raw-processed-outputs split is not a style preference; it is how you keep the audit trail intact when a number is challenged. If the raw tab has been edited in place, the chain is broken and the only honest answer is "I cannot prove where this came from." That is a career-limiting answer. Make it visceral by asking: "If your director emails you at 9 p.m. asking where the 47-minute MTTR came from, can you get it back to the raw export without guessing?"

Data Entry, Validation, and Cleanup

The big teachable moment in this section is the missing-severity decision. Students will reflexively want to delete the nineteen rows because it makes the spreadsheet "cleaner." Push back: "If you delete them and your boss later asks 'how many tickets did we actually get last month,' you will be wrong by nineteen." Flagging preserves both the count and the ability to filter. This is also the place to introduce the habit of

writing a short `README` tab in any workbook you share — listing the data sources, the cleanup steps, and any decisions like this one — so the next reader can reproduce the work.

On the mechanical side, make sure the class sees the difference between reformatting a cell (cosmetic only — underlying type unchanged) and using Text to Columns (actually converts the type). This is the source of the `#VALUE!` errors they will hit in the lab if they skip the correct step. Demonstrate the difference live with a single date cell on a throwaway sheet.

One caveat to name up front: when the lesson discusses Service Level Agreement (SLA) targets and the Key Performance Indicator (KPI) summary, students sometimes treat them as abstract. They are not — the SLA is a real promise OrbitCo makes to its customers (99.9% uptime; sub-60-minute resolution on High severity), and the KPI table is the evidence that leadership reads first every month.

Formulas for Technical Metrics

Walk through the MTTR and uptime calculations live, keystroke by keystroke, with the instructor's screen on the projector. Emphasize that MTTR in minutes is the right unit for operations reporting — a number like "MTTR: 47" is easier to reason about than "MTTR: 0.78 hours." The most common student error in this section is writing `XLOOKUP` with the `lookup_array` and `return_array` swapped; if a student's formula returns `#N/A` or the wrong value, check argument order first. The second most common is formatting: they get 0.999 and do not realize they need to change the cell format to percentage. Be ready for both.

Frame `XLOOKUP` as the modern replacement for `VLOOKUP`. If the class has seniors in it who learned `VLOOKUP` years ago, acknowledge it — then be explicit that `VLOOKUP`'s left-to-right and column-index constraints have caused enough production bugs over twenty years that the industry has moved on. Use `XLOOKUP` on new work.

Tables and Structured References

Convert a data range to a Table live on the projector so the class sees the Format as Table dialog and the auto-named `Table1`. Rename it to `Tickets` on camera so the habit sticks. Then type `=COUNTIF(Tickets[Severity], "High")` in a scratch cell and contrast it with the equivalent `=COUNTIF(B2:B421, "High")` — same answer today,

different behavior when a new row is pasted tomorrow. This is the single most durable reliability habit in the lesson, and the one that traces to the OJL 5d competency (Excel as living operations documentation).

Show the structured-reference autocomplete: when you type `Tickets[` in a formula, Excel offers the column list. That autocomplete is the argument for descriptive column headers — `ResolutionMinutes` beats `RM` because the formula reads like English.

Sorting, Filtering, and Conditional Formatting

Live-demo the SLA-breach conditional format on the OrbitCo tickets table. Start with no rule, scroll the table, and ask the class to find a breach row by eye. They will struggle. Add the rule; scroll again. The breach rows pop. The message — "let the tool do the work your eyes should not have to" — lands without needing to say it. The most common student mistake here is applying conditional formatting to a range rather than a Table column; when new rows are added, the rule does not extend. Show how applying it to the Table column (via structured reference or via "entire column of Table") keeps the rule alive as the data grows.

If time allows, also show the multi-column sort (Severity ascending, ResolutionMinutes descending) and the slicer UI (Insert → Slicer) — both are easy wins for the class's future dashboards.

Timing Suggestions

Segment	Time	Notes
Spreadsheet Design for Technical Data	6 min	Show the bad-workbook example; class lists problems
Data Entry, Validation, and Cleanup	7 min	Anchor the missing-severity decision; demo Text to Columns
Formulas for Technical Metrics	8 min	Live-calculate MTTR and uptime on the projector
Tables and Structured References	5 min	Convert range to Table on camera; rename; demo structured refs
Sorting, Filtering, and Conditional Formatting	5 min	Live-demo SLA-breach rule applied to the Table column
Activity: Clean the OrbitCo Incident Export	20 min	15 min individual work + 5 min debrief
Activity: Build the OrbitCo KPI Summary	15 min	12 min individual work + 3 min share-out
Wrap-Up and Reflection	4 min	Key takeaways; one reflection question live
Total	**60 min**	

Common Student Questions

Q: Do I really need to keep a `tickets_raw` sheet if the CSV file itself is still on disk? A: Yes. Weeks or months later, the CSV filename may drift, the folder may be reorganized, or the source system may have purged old exports. Keeping a raw tab inside the workbook binds the evidence to the analysis. It costs a few kilobytes and saves you from a "where did this number come from?" conversation you cannot answer.

Q: When should I use `SUMPRODUCT` instead of `COUNTIFS/SUMIFS`? A: Almost never in this course. `SUMPRODUCT` has a legitimate role in array math, but for the "count where severity is High and team is Platform" kind of question that dominates operations analysis, `COUNTIFS` and `SUMIFS` are more readable and auditable. If a colleague hands you a workbook full of `SUMPRODUCT` formulas, rewrite them to `COUNTIFS`/`SUMIFS` before you have to debug anything.

Q: What if the source system exports dates in a format Excel does not recognize at all? A: Use Text to Columns on the column with the "Date" option and pick the source format (MDY, DMY, or YMD). If that fails, the timestamps arrived as text with a non-standard separator; you will need `DATEVALUE` combined with `LEFT`/`MID`/`RIGHT` parsing, or a Power Query import. Escalate rather than guessing — wrong date parsing silently corrupts every downstream calculation.

Q: Is it ever acceptable to hardcode a cell reference instead of using a structured reference? A: Yes, for one-cell values that are not part of any Table — a lookup parameter, a reporting threshold, a constant like "30 days in minutes." Use named ranges for those so the formula still reads cleanly (`=1 - SUM(Uptime[DowntimeMinutes]) / MonthMinutes` rather than `=1 - SUM(Uptime[DowntimeMinutes]) / 43200`). Reserve raw cell addresses for throwaway scratch math.

Q: My conditional formatting rule is not updating when I paste in new rows. What did I do wrong? A: The rule was applied to a fixed range (e.g., `A2:G421`) rather than to the Table column. Open Home → Conditional Formatting → Manage Rules, delete the old rule, and re-apply it by first selecting the Table column header (which selects the entire Table column including future rows), then creating the rule. The "Applies to" field should read something like `=Tickets[Severity]` instead of `=$B$2:$B$421`.

Facilitation Tips

- **Open with the bad workbook, not with the definition.** Letting the class name the problems they already see (merged cells, embedded totals, color-as-data) activates prior knowledge and makes the "conventions" section feel like vocabulary, not rules.
- **Do not let a learner edit the raw sheet during the lab.** Watch for this specifically in the first five minutes of Activity 1. If someone cleans the raw sheet in place, stop them immediately and explain why — the lesson about evidence and audit trails is much harder to learn after the fact.
- **Type formulas on the projector, do not paste them.** Students need to see the structured-reference autocomplete happen to internalize that `Tickets[Severity]` is a real thing, not a magic string. Pasting skips the teaching moment.
- **Protect the debrief.** Both activities have short debrief windows (5 min and 3 min). Do not let group work run long at the expense of the debrief — the debrief is where the "why flag, not delete" and "why structured references" arguments solidify. Give clear time warnings.
- **Name the OJL competencies once, out loud, during the Tables section.** The OJL 2d and 5d tags on the activities are easy to skim past. Saying "this is the workbook you will still be maintaining two years from now — that is why the competency mapping calls this out" makes the connection to on-the-job practice explicit.